

Abstract Algebra

Sets & mathematical statements:

1: Equality Axiom

Two sets are equal only when they have the same elements

2: Empty Set Axiom:

There is a set called the "empty set", denoted as $\emptyset$, with no elements.

3: Pair Axiom

If A & B are sets, then $\{A, B\} = \{x \mid x=A \text{ or } x=B\}$ is a set. is a set 

4: Union Axiom

If S is a set of sets, then $\bigcup S = \bigcup_{A \in S} A = \{x \mid x \in A \text{ for some } A \in S\}$

5: Axiom of infinity

If we define the Natural numbers to be the sets $0 = \emptyset$, $1 = \{0\}$, $2 = \{0, 1\}$, $3 = \{0, 1, 2\}$, $\dots$, then $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ is a set.

6: Replacement Axiom

If A is a set & $F(x)$ is a statement about x and y with the property $\forall x \in A \exists! y (F(x, y))$, then $F(A)$ is a set, which $F(x, y)$ is true,

6: Replacement Axiom

If A is a set & $F(x)$ is a statement about x and y with the property that for every set x , $\exists$ a unique set $y = F(x)$ for which $F(x, y)$ is true, then $\{F(x) | x \in A\} = \{y | \exists x \in A F(x, y)\}$ is a set.

7: Separation Axiom

If A is a set & $F(x)$ is a statement about x , $\{x \in A | F(x)\}$ is a set.

8: Axiom of Choice

Given a nonempty set of non-empty disjoint sets, $\exists$ set C which contains exactly one element from each of the sets in S .

9: Power Set Axiom

If A is a set then $P(A) = \{X | X \subseteq A\}$ is a set, called the "Power set" of A .

Implications of set axioms:

1: Sets are determined by their contents, if two different sets have the same elements, the sets are equal.

2: Mathematics needs a starting element, with the empty set axiom, we can define more complex sets like the natural number system.

3: Any two existing sets can be packaged in as a single set.

4: $\bigcup S = \{1, 2, 3, 4\}$ when $S = \{\{1, 2\}, \{3, 4\}\}$

you can collapse a set of sets into the elements of the nested sets.

